

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA0713 COURSE

NAME: COMPUTER NETWORKS

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques

using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I **T.GANESH(192425246)**(Name / Reg No) certify that this submission is my original work

and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

T.GANESH

Signature of the student:

1) Suppose we want to transmit the message 11100011 and protect it from errors using the

CRC polynomial X^3+1 .

i.) Use polynomial long division to determine the message that should be transmitted.

ii) Suppose the leftmost bit of the message is inverted due to noise on the transmission

link. What is the result of the receiver's CRC calculation? How does the receiver know that

an error has occurred?

Aim

To determine the CRC bits for the given message using the generator polynomial (x^3+1) and verify whether an error introduced during transmission can be detected by the receiver.

Problem Understanding

In digital communication, data transmitted through a communication channel may be corrupted due to noise or interference. To detect such errors, the **Cyclic Redundancy Check (CRC)** technique is widely used. CRC treats the message and the generator polynomial as binary polynomials and performs modulo-2 division. The remainder obtained from the division is appended to the original message before transmission. At the receiver, the same division is performed. If the remainder is zero, the received message is assumed to be error-free; otherwise, an error is detected.

In this problem, the given message is **11100011**, and the generator polynomial is (x^3+1) . We must determine the transmitted message using CRC and then analyze what happens if the leftmost bit is inverted during transmission.

Given Data

Parameter	Value
Original Message	11100011
Generator Polynomial	(x^3+1)
Generator (Binary)	1001
Degree of Polynomial	3

Theory

The generator polynomial

$$\begin{aligned} &[\\ G(x) &= x^3 + 1 \\ &] \end{aligned}$$

is represented in binary form as

1001

Since the degree of the polynomial is **3**, three zeros must be appended to the original message before performing CRC division.

Original Message

11100011

Message after appending zeros

11100011000

Method Used

The CRC algorithm consists of the following steps:

1. Append three zeros to the original message.
2. Divide the resulting bit sequence by the generator polynomial using **Modulo-2 (XOR) division**.
3. Obtain the remainder.
4. Append the remainder to the original message.
5. The resulting bit sequence is transmitted.
6. At the receiver, divide the received message by the same generator polynomial.
7. If the remainder is **000**, the transmission is correct; otherwise, an error has occurred.

Solution

Step 1: Append Zeros

Original Message

11100011

Appending three zeros

11100011000

Step 2: Perform CRC Division

Generator

1001

Dividend

11100011000

Perform modulo-2 division (XOR operation) until the complete message is processed.

After completing the division, the remainder obtained is

011

Step 3: Generate the Transmitted Message

Original Message

11100011

CRC Remainder

011

Hence, the transmitted codeword is

11100011011

Answer for Part (i)

CRC Remainder

011

Message to be Transmitted

11100011011

Part (ii): Error Detection

Assume that noise in the communication channel changes the **leftmost bit** of the transmitted message.

Original transmitted message

11100011011

Received message after the error

01100011011

The receiver again performs CRC division using the same generator polynomial (**1001**).

Since one bit has changed, the received message is **not exactly divisible** by the generator polynomial.

Therefore,

Remainder $\neq$ 000

A non-zero remainder indicates that an error has occurred during transmission.

How the Receiver Detects the Error

The receiver checks the remainder after performing CRC division.

Remainder Interpretation

000 No error detected

Non-zero Error detected

Since the received message produces a **non-zero remainder**, the receiver immediately concludes that the frame has been

corrupted and rejects it (or requests retransmission, depending on the protocol).

Result

- CRC Generator = **1001**
- CRC Remainder = **011**
- Transmitted Codeword = **11100011011**
- Error introduced by changing the first bit.
- Receiver obtains a **non-zero remainder**.
- Hence, the transmission error is successfully detected.

Interpretation of Results

CRC is one of the most reliable error-detection techniques used in computer networks. It is implemented in technologies such as **Ethernet, Wi-Fi, USB, hard disks, and communication protocols** to detect accidental bit errors during transmission. In this problem, the CRC mechanism successfully detects the single-bit error because the received codeword no longer divides evenly by the generator polynomial. This ensures that corrupted data is not accepted by the receiver, thereby improving communication reliability.

Final Answer

Item	Result
Original Message	11100011
Generator Polynomial	(x^3+1)

Binary Generator	1001
CRC Remainder	011
Transmitted Message	11100011011
Receiver Remainder after Error	Non-zero

Question 2

a) A network with a bandwidth of **10 Mbps** can pass only an average of **12,000 frames per minute**, with each frame carrying an average of **10,000 bits**. What is the **throughput** of this network?

b) A network has a bandwidth of **3000 Hz (300 Hz to 3300 Hz)** assigned for data communications. The **signal-to-noise ratio (SNR)** is **3162**. Find the **channel capacity**.

Aim

To calculate the **throughput** of a network based on the number of frames transmitted and to determine the **maximum channel capacity** using the Shannon Capacity Theorem.

Problem Understanding

In computer networks, **bandwidth** refers to the maximum data transmission capacity of a communication channel, whereas **throughput** represents the actual amount of data successfully transmitted over the network in a given period of time.

Even though a network may have a high bandwidth, the actual throughput may be lower due to factors such as network congestion, protocol overhead, retransmissions, and transmission delays.

The second part of the problem uses **Shannon's Channel Capacity Theorem**, which determines the maximum theoretical data rate that can be achieved over a noisy communication channel.

Given Data

Question 2(a)

Parameter	Value
Network Bandwidth	10 Mbps
Frames per Minute	12,000
Bits per Frame	10,000 bits

Question 2(b)

Parameter	Value
Bandwidth (B)	3000 Hz
Signal-to-Noise Ratio (SNR)	3162

Theory

Throughput

Throughput is the amount of useful data successfully transmitted through a communication channel in one second.

Formula

$$\left[\begin{array}{l} \text{Throughput} = \frac{\text{Total Data Transmitted}}{\text{Time}} \end{array} \right]$$

Shannon Channel Capacity

The Shannon Capacity Theorem gives the maximum theoretical data transmission rate over a noisy channel.

Formula

$$\left[\begin{array}{l} C = B \log_2(1 + \text{SNR}) \end{array} \right]$$

Where,

- **C** = Channel Capacity (bps)
- **B** = Channel Bandwidth (Hz)
- **SNR** = Signal-to-Noise ratio

Solution

Question 2(a): Calculate the Throughput

Step 1: Calculate Total Data Transmitted in One Minute

Number of Frames

= 12,000 frames

Bits in each Frame

= 10,000 bits

Therefore,

Total Data

= $12,000 \times 10,000$

= **120,000,000 bits per minute**

Step 2: Convert Data Per Minute to Data Per Second

There are **60 seconds** in one minute.

$$\left[\frac{\text{Throughput}}{\frac{120,000,000}{60}} \right]$$
$$\left[= 2,000,000 \text{ bits/second} \right]$$

Therefore,

Throughput = 2 Mbps

Step 3: Calculate Network Efficiency

Given Network Bandwidth

= 10 Mbps

Calculated Throughput

= 2 Mbps

$$\left[\frac{\text{Efficiency}}{\frac{2}{10}} \times 100 \right]$$

= **20%**

Answer for Question 2(a)

Parameter	Value
Bandwidth	10 Mbps
Actual Throughput	2 Mbps
Network Efficiency	20%

Solution

Question 2(b): Calculate Channel Capacity

Step 1: Write Shannon's Formula

$$C = B \log_2(1 + \text{SNR})$$

Where,

B = 3000 Hz

SNR = 3162

Step 2: Substitute the Values

$$[C = 3000 \log_2(1 + 3162)]$$

$$[= 3000 \log_2(3163)]$$

Since

$$[\log_2(3163) \approx 11.63]$$

Therefore,

$$[C = 3000 \times 11.63]$$

$$[= 34,890 \text{ bps}]$$

Step 3: Convert into kbps

$$[34,890 \text{ bps}]$$

$$[34.89 \text{ kbps}]$$

Answer for Question 2(b)

Parameter	Value
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Bandwidth	3000 Hz
SNR	3162
Channel Capacity	34,890 bps
In kbps	34.89 kbps

Result

Question 2(a)

- Total Data Transmitted = **120,000,000 bits/minute**
- Throughput = **2 Mbps**
- Network Efficiency = **20%**

Question 2(b)

- Channel Capacity = **34,890 bps**
- Equivalent to **34.89 kbps**

Interpretation of Results

The calculated throughput of **2 Mbps** indicates that although the network supports a maximum bandwidth of **10 Mbps**, only **20%** of its capacity is being utilized. This reduction may be caused by network congestion, transmission overhead, retransmissions, or protocol inefficiencies.

Using Shannon's theorem, the maximum theoretical channel capacity for the given noisy communication channel is **34.89 kbps**. This represents the highest possible data rate that can be achieved without significant transmission errors under the given bandwidth and signal-to-noise ratio.

Conclusion

The network throughput was calculated as **2 Mbps**, which is lower than the available bandwidth due to practical network limitations. Using Shannon's Channel Capacity Theorem, the maximum theoretical data transmission rate for the given communication channel was found to be **34.89 kbps**. These calculations demonstrate the difference between **actual network performance (throughput)** and **theoretical communication limits (channel capacity)**, both of which are essential concepts in computer networks.

Question 3

A multispecialty hospital uses **wireless patient-monitoring devices** to continuously transmit patients' heart rate and blood pressure readings to a central monitoring station. Since the data is transmitted over a wireless channel, **electromagnetic interference occasionally changes one bit during transmission**. To ensure that doctors receive accurate patient information without waiting for retransmission, the hospital implements the **Hamming (12,8) error-correcting code**.

Explain how the **Hamming (12,8) code** is used to detect and correct a single-bit error during transmission.

Aim

To understand the working of the **Hamming (12,8) Error-Correcting Code** and demonstrate how it detects and corrects a single-bit error in transmitted patient-monitoring data.

Problem Understanding

In modern hospitals, wireless patient-monitoring devices continuously send vital health information, such as heart rate and blood pressure, to a central monitoring station. Since wireless communication is affected by electromagnetic interference, transmitted data may become corrupted.

If a single bit changes during transmission, incorrect patient information could be displayed, leading to wrong medical decisions. To avoid this problem, the hospital uses the **Hamming (12,8) code**, which can detect and automatically correct a single-bit error without requesting retransmission.

This improves the reliability and speed of healthcare communication systems.

Given Data

Parameter	Value
Error Control Technique	Hamming (12,8) Code
Data Bits	8
Total Bits after Encoding	12
Parity Bits	4
Error Type	Single-Bit Error

Theory

The **Hamming (12,8)** code is an error-correcting code used in digital communication systems. It converts **8 data bits** into a **12-bit codeword** by adding **4 parity bits**.

The parity bits are placed at positions that are powers of 2.

Position	Bit Type
1	P1
2	P2
3	D1
4	P4
5	D2
6	D3
7	D4
8	P8
9	D5
10	D6
11	D7
12	D8

Where,

- **P1, P2, P4, and P8** are parity bits.
- **D1 to D8** are data bits.

These parity bits help identify the position of an error during transmission.

Formula

The number of parity bits required is calculated using:

$$\left[2^r \geq m+r+1 \right]$$

Where,

- (m) = Number of data bits
- (r) = Number of parity bits

For this problem,

$$\left[m=8 \right]$$

Trying (r=4)

$$\left[2^4=16 \right]$$

$$\left[8+4+1=13 \right]$$

Since

$$\left[16 \geq 13 \right]$$

Therefore,

4 parity bits are required.

Solution

Assume the hospital wants to transmit the following **8-bit patient data**.

Data Bits

10110011

Step 1: Place Data Bits

Insert the data bits into the non-parity positions.

Position	1	2	3	4	5	6	7	8	9	10	11	12
Bit Type	P	P	D	P	D	D	D	P	D	D	D	D
Value	?	?	1	?	0	1	1	?	0	0	1	1

Step 2: Calculate the Parity Bits

Using **Even Parity**:

- **P1** checks positions **1, 3, 5, 7, 9, 11**
- **P2** checks positions **2, 3, 6, 7, 10, 11**
- **P4** checks positions **4, 5, 6, 7, 12**
- **P8** checks positions **8, 9, 10, 11, 12**

After calculation,

Parity Bit	Value
P1	1
P2	0

P4	1
P8	0

Step 3: Construct the Hamming Code

The transmitted **12-bit codeword** becomes

101101100011

Step 4: Error During Transmission

Assume electromagnetic interference changes **Bit 7**.

Original Codeword

101101100011

Received Codeword

101101000011

Step 5: Receiver Checks the Parity

The receiver recalculates all parity bits.

The syndrome bits obtained are

0111

Binary

$0111_2 = 7_{10}$

Therefore,

Error Position = Bit 7

The receiver flips Bit 7 back to its original value.

Corrected Codeword

101101100011

The original patient data is recovered successfully.

Final Answer

Item	Value
Data Bits	10110011
Parity Bits	4
Encoded Bits	12
Error Position	Bit 7
Correction	Bit 7 is inverted
Recovered Data	10110011

Result

The **Hamming (12,8) code** successfully detected and corrected the single-bit error introduced during wireless transmission. The receiver identified the error location using the syndrome bits and restored the original patient data without requesting retransmission.

Interpretation of Results

Hamming codes are widely used in applications where immediate and reliable communication is essential. In the hospital scenario, the Hamming (12,8) code ensures that patient health data remains accurate even if a single-bit error occurs during wireless transmission. This avoids delays caused by

retransmissions and enables doctors to receive correct patient information in real time.

Conclusion

The **Hamming (12,8) error-correcting code** is an effective technique for ensuring reliable wireless communication. By adding four parity bits to eight data bits, it can detect and automatically correct any single-bit error. This makes it highly suitable for healthcare monitoring systems, satellite communication, memory systems, and other applications where data accuracy is critical.

Question 4

4(a)

A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments.

Using the given subnet mask:

- 1. Calculate the total number of subnets created.**
- 2. Calculate the number of usable hosts per subnet.**
- 3. Determine the subnet to which the IP address 172.16.10.78 belongs.**
- 4. Identify the network address and broadcast address for that subnet.**
- 5. Determine whether the IP address 172.16.10.95 falls within the same subnet range.**

4(b)

A network uses the IPv6 address:

2001:0db8:0000:0000:0000:ff00:0042:8329

Using IPv6 addressing rules:

- 1. Compress the given IPv6 address into its shortest form.**
- 2. Expand the compressed address back into its full notation.**
- 3. Using a /64 prefix length, determine the network prefix.**
- 4. Calculate the total number of possible host addresses available within this network.**

Aim

To understand IPv4 subnetting and IPv6 addressing by calculating subnets, host addresses, network information, and IPv6 compression and expansion.

Problem Understanding

Subnetting is the process of dividing a large network into smaller logical networks. It improves network performance, security, and efficient utilization of IP addresses.

IPv6 is the next-generation Internet Protocol that provides a much larger address space than IPv4. IPv6 allows address compression and uses prefix lengths to identify network and host portions.

Given Data

Question 4(a)

Parameter	Value
Network Address	172.16.10.0
Original Prefix	/24
New Prefix	/27

Question 4(b)

Parameter	Value
IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Prefix Length	/64

Theory

IPv4 Subnetting

Subnetting divides a network into smaller subnetworks by borrowing bits from the host portion.

Formula for Number of Subnets

[
Number of Subnets $\}=2^n$
]

Where

- **n = Borrowed Bits**

Formula for Number of Hosts

$$\left[\frac{\text{Usable Hosts}}{2} \right] = 2^{h-2}$$

Where

- **h = Host Bits**

The subtraction of **2** is because one address is reserved as the **Network Address** and one as the **Broadcast Address**.

IPv6 Addressing

IPv6 addresses contain **128 bits** and are written as **8 groups** of four hexadecimal digits.

Rules:

- Leading zeros can be removed.
- One continuous sequence of zero groups can be replaced by ::.

Solution

Question 4(a)

Step 1: Calculate Borrowed Bits

Original Prefix

/24

New Prefix

/27

Borrowed Bits

$$[$$

$$27-24=3$$

$$]$$

Step 2: Calculate Number of Subnets

Using the formula

$$[\text{Number of Subnets}] = 2^n$$

$$[$$

$$=2^3$$

$$]$$

$$[$$

$$=8$$

$$]$$

Total Number of Subnets = 8

Step 3: Calculate Usable Hosts

Host Bits

$$[$$

$$32-27=5$$

$$]$$

Using the formula

$$[$$

$$2^5-2$$

$$]$$

[
32-2=30
]

Usable Hosts per Subnet = 30

Step 4: List the Subnets

Since each subnet contains **32 addresses**, the subnet ranges are:

Subnet	Address Range
1	172.16.10.0 – 172.16.10.31
2	172.16.10.32 – 172.16.10.63
3	172.16.10.64 – 172.16.10.95
4	172.16.10.96 – 172.16.10.127
5	172.16.10.128 – 172.16.10.159
6	172.16.10.160 – 172.16.10.191
7	172.16.10.192 – 172.16.10.223
8	172.16.10.224 – 172.16.10.255

Step 5: Find the Subnet for 172.16.10.78

The IP Address

172.16.10.78

falls in the range

172.16.10.64 – 172.16.10.95

Therefore,

Subnet = 172.16.10.64/27

Step 6: Determine Network Address

Network Address

172.16.10.64

Step 7: Determine Broadcast Address

Broadcast Address

172.16.10.95

Step 8: Check Whether 172.16.10.95 Belongs to the Same Subnet

The subnet range is

172.16.10.64 – 172.16.10.95

Given IP

172.16.10.95

This is the **Broadcast Address** of the subnet.

Therefore,

Yes, 172.16.10.95 belongs to the same subnet, but it is reserved as the Broadcast Address and cannot be assigned to a host.

Answer for Question 4(a)

Parameter	Answer
Total Subnets	8
Usable Hosts/Subnet	30
Subnet of 172.16.10.78	172.16.10.64/27
Network Address	172.16.10.64
Broadcast Address	172.16.10.95
Does 172.16.10.95 belong to same subnet?	Yes (Broadcast Address)

Solution

Question 4(b)

Step 1: Compress the IPv6 Address

Given Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Remove leading zeros

2001:db8:0:0:0:ff00:42:8329

Replace consecutive zero groups with ::

2001:db8::ff00:42:8329

Compressed IPv6 Address

2001:db8::ff00:42:8329

Step 2: Expand the IPv6 Address

Expanded Form

2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Determine the Network Prefix

Given Prefix Length

/64

The first **64 bits** represent the network.

Therefore,

Network Prefix

2001:0db8:0000:0000::/64

Step 4: Calculate Host Addresses

Host Bits

[
128-64=64
]

Possible Host Addresses

[
 $2^{\{64\}}$
]

[
=18,446,744,073,709,551,616
]

Approximately

18.4 Quintillion Host Addresses

Answer for Question 4(b)

Parameter	Answer
Compressed IPv6 Address	2001:db8::ff00:42:8329
Expanded IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix	2001:0db8:0000:0000::/64
Possible Host Addresses	$(2^{\{64\}}) = 18,446,744,073,709,551,616$

Result

IPv4

- Total Subnets = **8**
- Usable Hosts per Subnet = **30**
- Subnet for **172.16.10.78** = **172.16.10.64/27**
- Network Address = **172.16.10.64**
- Broadcast Address = **172.16.10.95**
- IP Address **172.16.10.95** belongs to the same subnet but is reserved as the broadcast address.

IPv6

- Compressed Address = **2001:db8::ff00:42:8329**

- Expanded Address = **2001:0db8:0000:0000:0000:ff00:0042:8329**
- Network Prefix = **2001:0db8:0000:0000::/64**
- Total Possible Host Addresses = **2^{64} = 18,446,744,073,709,551,616**

Interpretation of Results

IPv4 subnetting divides a larger network into multiple smaller networks, improving network organization, security, and efficient address allocation. In this example, a **/24 network** was divided into **8 subnets**, each supporting **30 usable hosts**.

IPv6 provides a much larger address space than IPv4. With a **/64 prefix**, the remaining **64 bits** are available for host addresses, allowing approximately **18.4 quintillion unique host addresses** within a single subnet. This vast address space ensures scalability for modern Internet and IoT applications.

Conclusion

This problem demonstrates the practical implementation of **IPv4 subnetting** and **IPv6 addressing**. IPv4 subnetting enables efficient utilization of available IP addresses by creating smaller subnets, while IPv6 offers an enormous address space through 128-bit addressing. Both techniques are essential for designing scalable, secure, and efficient computer networks.

Question 5

A university is conducting an **online semester examination** for **8,000 students** simultaneously. At exactly **10:00 AM**, the examination server begins transmitting encrypted question papers to all students. The university server is connected through a **10 Gbps high-speed network**, while students access the examination portal using different internet connections such as **fiber broadband (100 Mbps)**, **home Wi-Fi (20 Mbps)**, and **mobile networks (5–10 Mbps)**.

During the first few minutes of the examination, many students with slower internet connections report **incomplete downloads, repeated retransmissions, delayed acknowledgments, and temporary freezing of the examination portal**. Network administrators observe that the server continues transmitting data at a high rate even though several student devices are unable to receive and process packets at the same speed. This results in **receiver buffer overflow, unnecessary retransmissions, increased network traffic, and delays in delivering the question papers**.

Analyze the above scenario and identify the primary networking problem. Explain how the mismatch between the sender's transmission speed and the receiver's processing capability affects communication.

Aim

To analyze the impact of **flow control problems** in a communication network and explain how a mismatch between the sender's transmission speed and the receiver's processing capability affects reliable data transmission.

Problem Understanding

In any communication network, the sender and receiver may have different transmission and processing speeds. If the sender transmits data faster than the receiver can process it, the receiver's buffer becomes full, leading to data loss and retransmissions.

In this scenario, the university examination server is connected to a **10 Gbps** high-speed network, whereas students use internet connections ranging from **5 Mbps to 100 Mbps**. Because the server sends encrypted question papers at a much higher rate than many student devices can receive, communication problems occur. This situation highlights the importance of **Flow Control** in the Transport Layer of the OSI model.

Given Data

Parameter	Value
Number of Students	8,000
Server Speed	10 Gbps
Student Internet Speed	5 Mbps – 100 Mbps
Problems Observed	Incomplete downloads, retransmissions, delayed ACKs, buffer overflow

Theory

Flow Control

Flow Control is a mechanism used in the **Transport Layer** to regulate the rate of data transmission between the sender and receiver. Its primary objective is to ensure that the sender does not transmit data faster than the receiver can accept and process.

Flow Control improves communication reliability by preventing:

- Receiver buffer overflow
- Packet loss
- Unnecessary retransmissions
- Network congestion
- Delayed acknowledgments

Common flow control techniques include:

- Stop-and-Wait Protocol
- Sliding Window Protocol
- TCP Flow Control (Receiver Window)

Solution

Step 1: Identify the Primary Networking Problem

The primary networking problem in this scenario is:

Flow Control Failure

The examination server transmits data at **10 Gbps**, while many students receive data at speeds between **5 Mbps and 100 Mbps**.

Since the sender is significantly faster than the receivers, many student devices cannot process incoming packets quickly enough.

Step 2: Analyze the Speed Mismatch

Server Transmission Speed

10 Gbps

Student Internet Speeds

- Fiber Broadband = **100 Mbps**
- Home Wi-Fi = **20 Mbps**
- Mobile Network = **5–10 Mbps**

The server sends data much faster than the receivers can process.

This difference causes communication imbalance.

Step 3: Effect on Receiver Buffer

As packets arrive continuously,

- The receiver stores packets in its **buffer memory**.
- Slow receivers cannot process packets quickly.
- The buffer gradually becomes full.
- Once the buffer is full, new packets are discarded.

This condition is known as:

Receiver Buffer Overflow

Step 4: Consequences of Buffer Overflow

Because packets are dropped,

- Students receive incomplete question papers.
- Missing packets trigger retransmissions.
- Network traffic increases.
- Additional bandwidth is consumed.
- Students experience delays in downloading question papers.

Step 5: Delayed Acknowledgments

Since student devices process packets slowly,

- Acknowledgment (ACK) packets are delayed.
- The server assumes packets were lost.
- The server retransmits the same packets.
- Duplicate packets increase network congestion.

Step 6: Overall Communication Problems

The mismatch between sender and receiver speeds results in the following issues:

Problem	Effect
Buffer Overflow	Receiver cannot store incoming packets
Packet Loss	Some packets are discarded

Retransmissions	Same packets are sent again
Delayed ACKs	Sender waits longer for confirmation
Network Congestion	Extra traffic reduces performance
Download Delay	Students receive question papers late
Temporary Freezing	Examination portal becomes unresponsive

Step 7: Suggested Solution

The university should implement **Flow Control** mechanisms such as:

Sliding Window Protocol

- Controls the number of packets sent before receiving acknowledgments.
- Prevents receiver buffer overflow.
- Improves transmission efficiency.

TCP Flow Control

TCP automatically adjusts the transmission speed based on the receiver's available buffer size.

Congestion Control

The server should gradually increase its transmission rate instead of sending all packets simultaneously.

Load Balancing

Multiple servers can be used to distribute traffic among students, reducing network congestion.

Final Answer

The primary networking problem in the given scenario is **Flow Control Failure**.

The examination server transmits encrypted question papers at a much higher speed (**10 Gbps**) than many students can receive and process (**5–100 Mbps**). This speed mismatch causes the receiver's buffer to overflow, resulting in packet loss, repeated retransmissions, delayed acknowledgments, increased network traffic, and slower delivery of question papers.

Implementing **Sliding Window Protocol**, **TCP Flow Control**, **Congestion Control**, and **Load Balancing** can effectively solve these problems by matching the sender's transmission rate with the receiver's processing capability.

Result

The analysis shows that a mismatch between the sender's transmission speed and the receiver's processing capability leads to **Flow Control Failure**. This causes receiver buffer overflow, packet loss, retransmissions, delayed acknowledgments, network congestion, and slower communication. Proper implementation of flow control mechanisms ensures reliable and efficient data transmission.

Interpretation of Results

This scenario demonstrates the practical importance of **Flow Control** in modern computer networks. High-speed servers must regulate their transmission rates according to the capacity of receiving devices. Techniques such as the **Sliding Window Protocol**, **TCP Flow Control**, and **Congestion Control** help maintain smooth communication by preventing buffer overflow and reducing unnecessary retransmissions. These mechanisms are especially critical in real-time applications such as online examinations, video conferencing, cloud services, and online banking, where reliable and timely data delivery is essential.

Conclusion

The university's online examination system experienced communication problems because the server transmitted data much faster than many students' devices could receive and process. This mismatch caused **Flow Control Failure**, leading to receiver buffer overflow, packet loss, retransmissions, and increased network congestion. By implementing appropriate flow control and congestion control techniques, the university can ensure that encrypted question papers are delivered reliably, efficiently, and without delays to all students, regardless of their internet connection speed.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation